Sci30105 Modern Physics

Special Relativity

Lect 1: Origin of Special Relativity

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Outline of Special Relativity

- Origin of special relativity
- Consequence of Einstein' theory
- The Lorentz transformation
- Relativity of momentum and energy
- General relativity

Leaning Objectives

By the end of this section, you will be able to

- Describe the difference between inertial and non-inertial frame

- Describe the problem of classical mechanic (Galilean Relativity)

Relativity (สัมพัทธภาพ)

Relativity (สัมพัทธภาพ)

The concept that description and measurement of physics quantities

such as position,

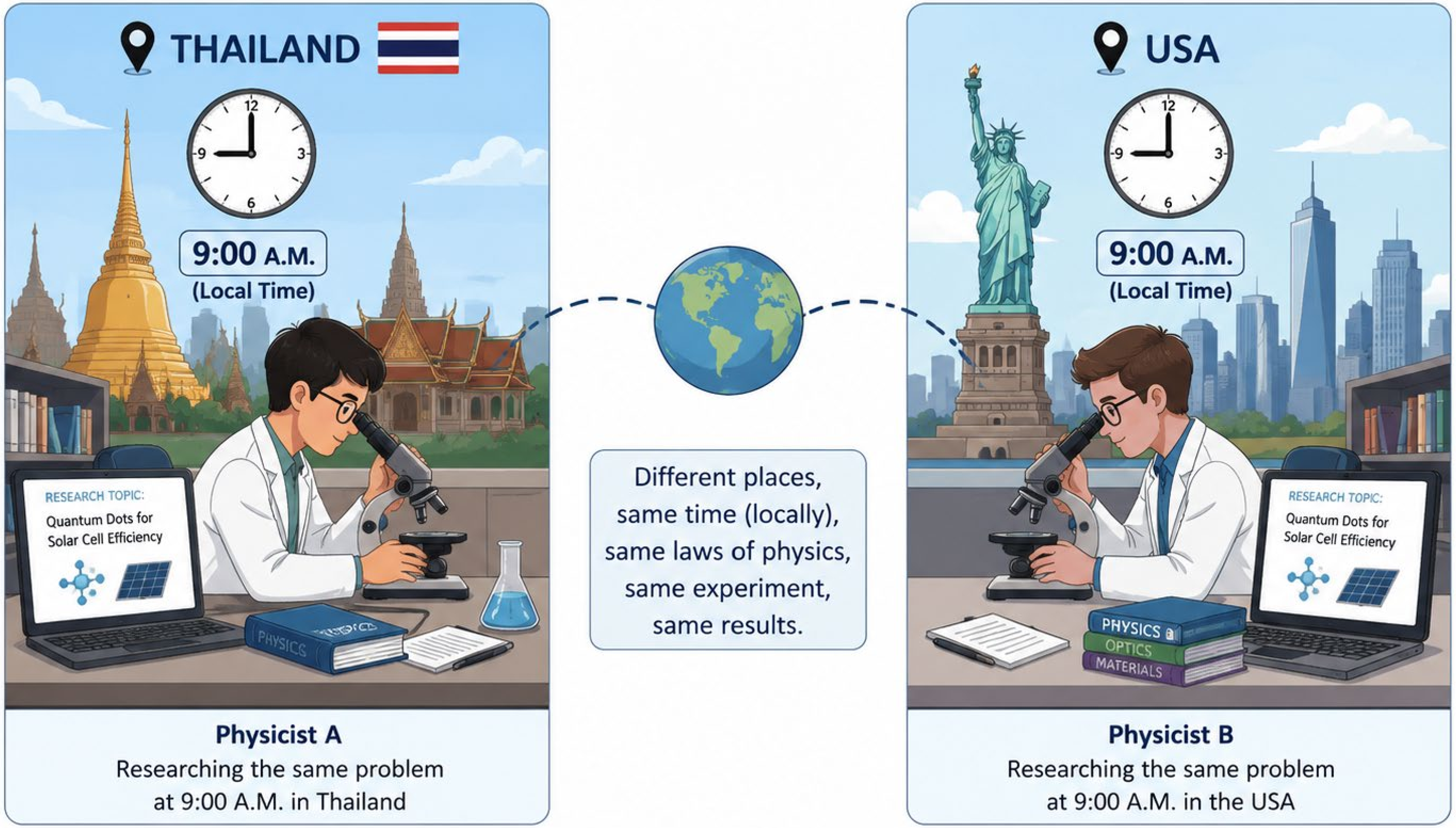
distance,

time of event,

and velocity

may differ depending on the motion and viewpoint of **the observer**

Principle of Relativity :



Adopted from: ChatGPT Image (2026)

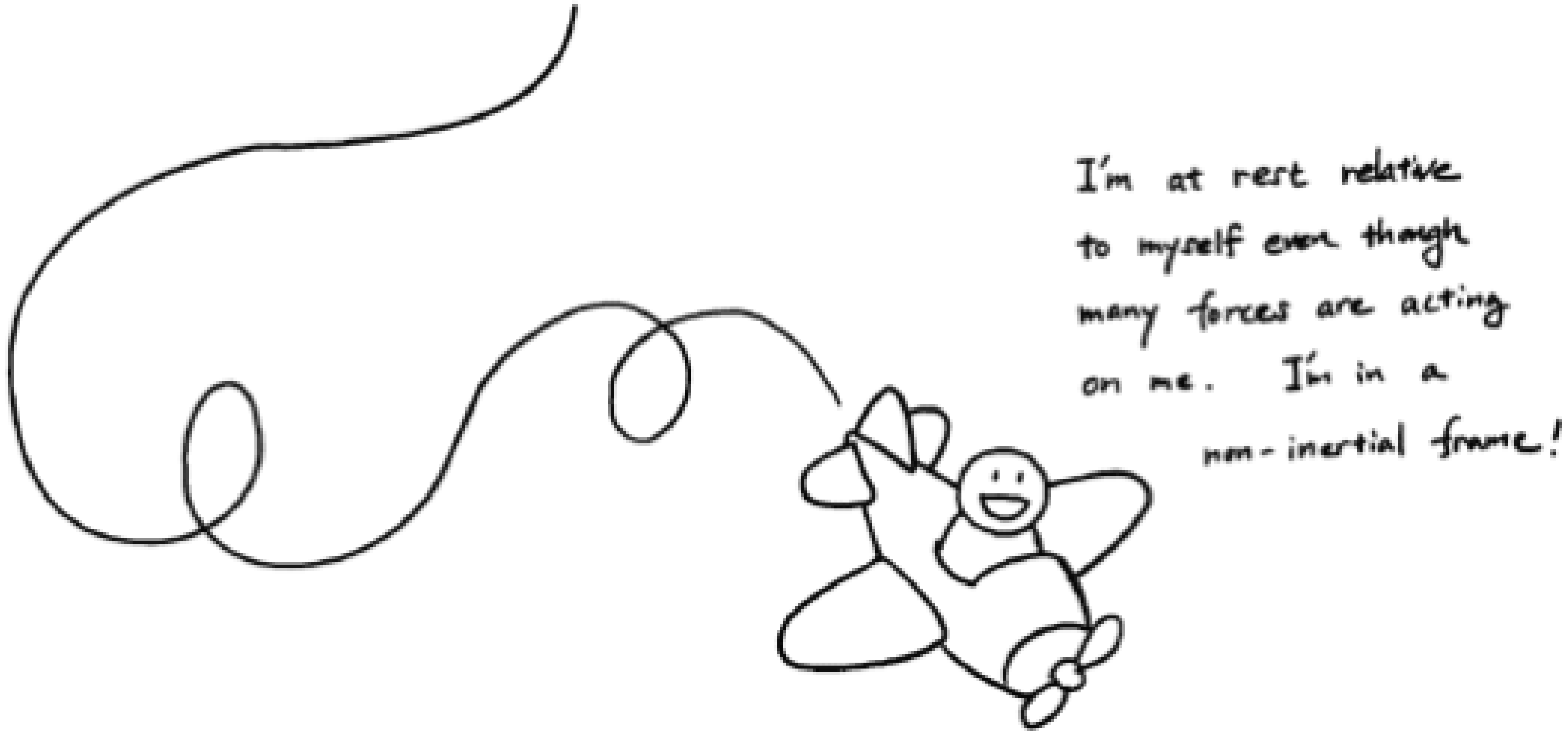
Galilean relativity :

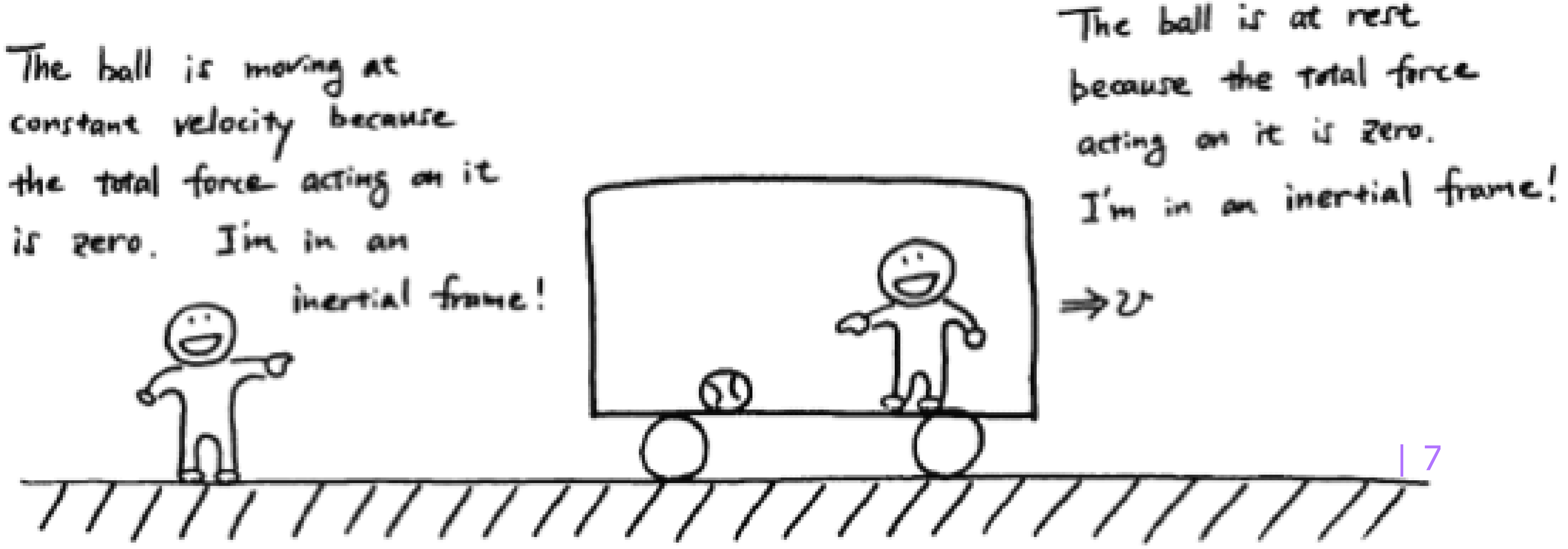
"All **inertial observers** describe the laws of physics in the same way,

even though they may measure different velocities. (in absolute time)"

The law of Physics must be the same in all inertial frame of reference

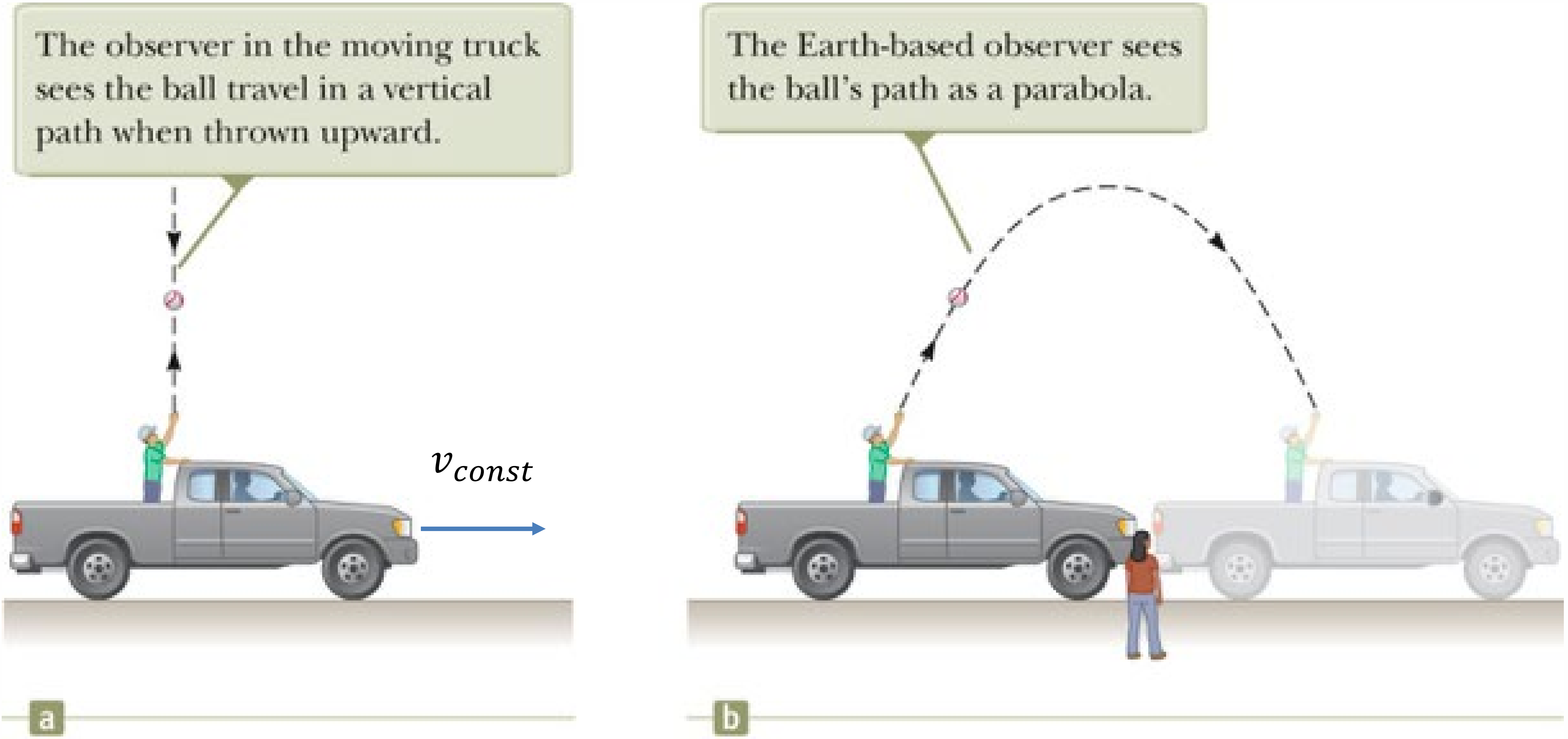
"Newton's law is correct when we assume that it are inertial frame"





Adapted from: T. Takuechi, Physics Dept., College of Science, Virginia Tech.

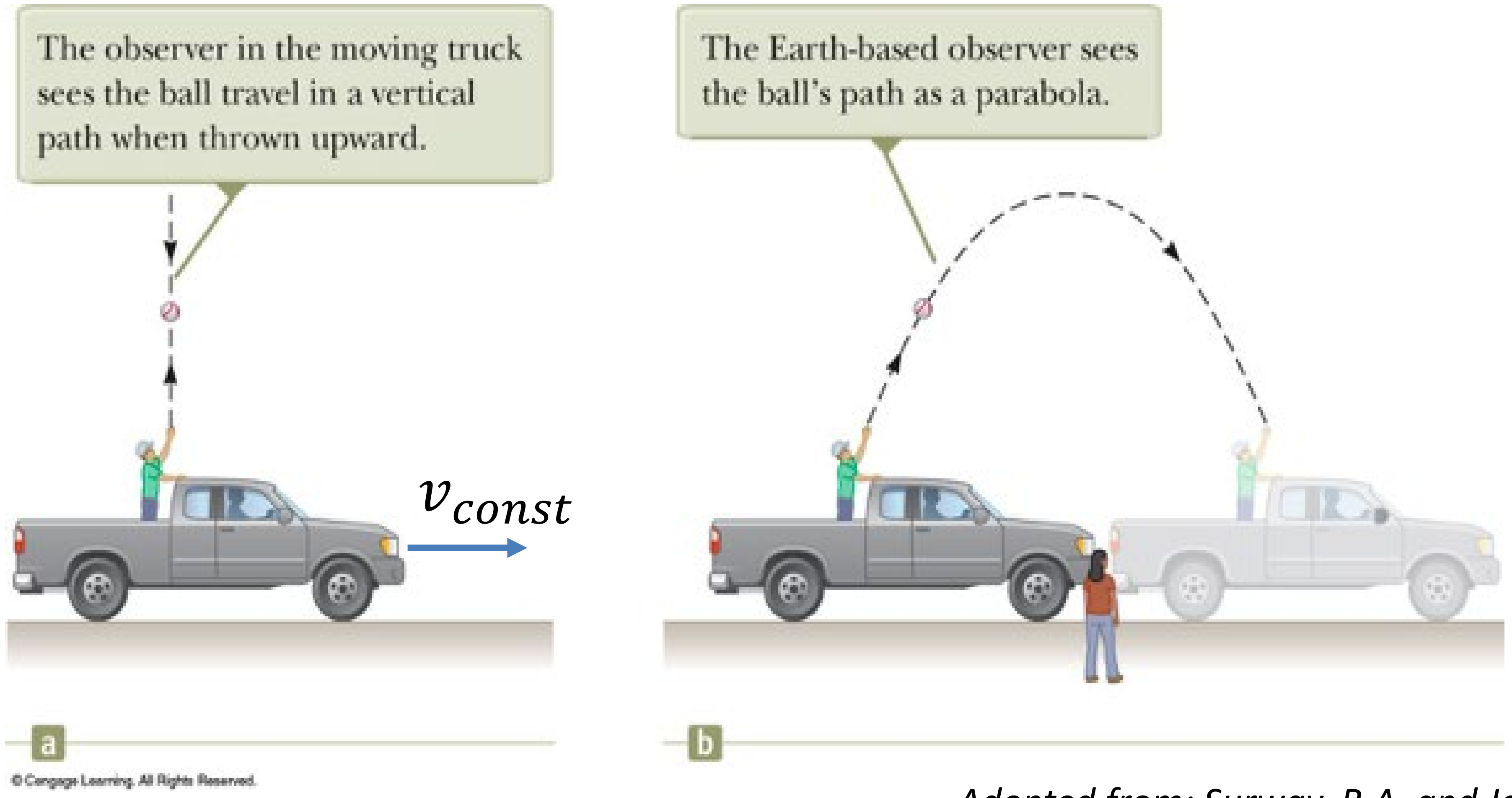
Galilean Relativity (Example)



v_{const}

Adopted from: Surway, R.A. and Jewett, J.W. (2014)

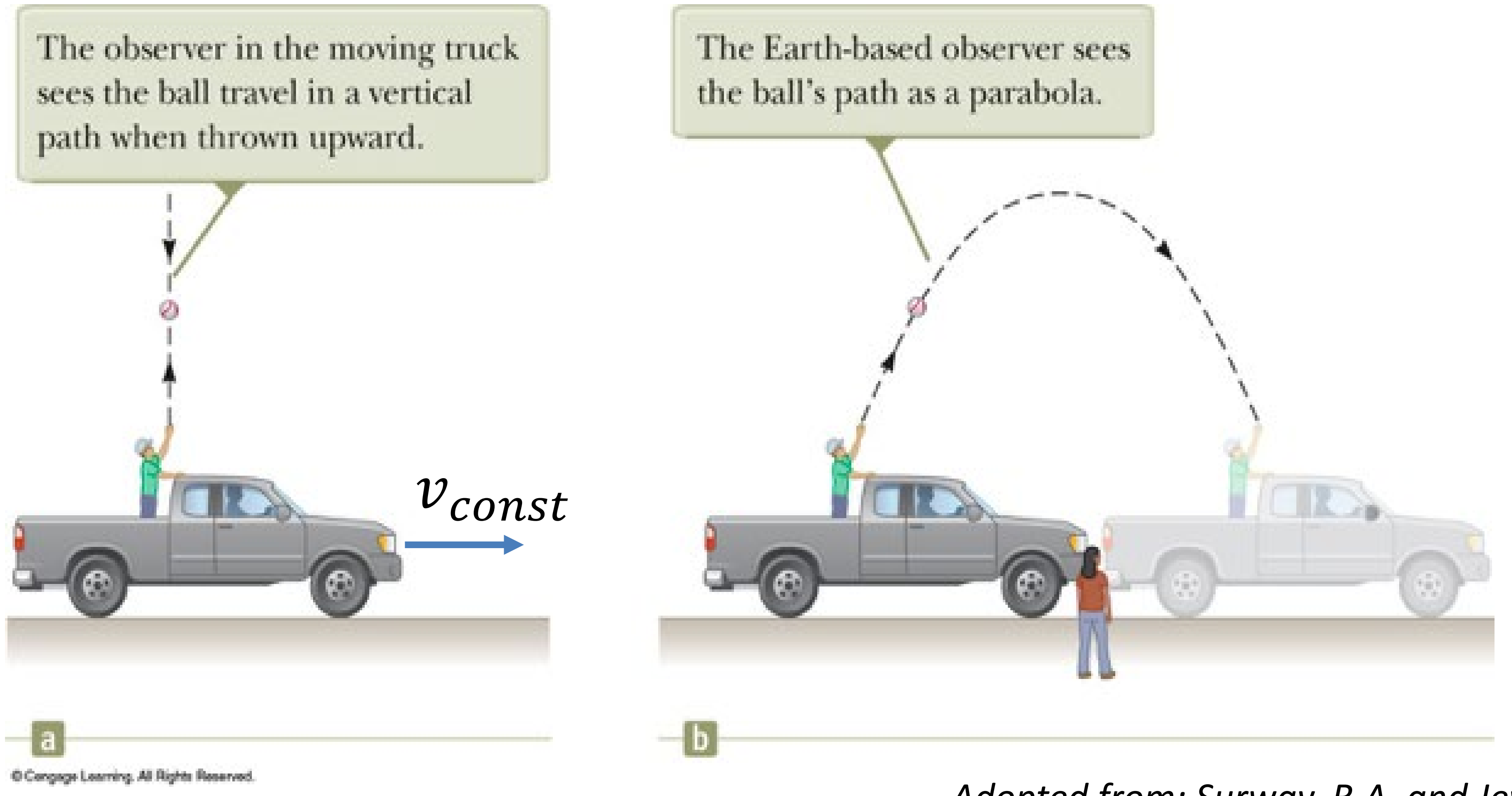
Galilean relativity (Example)



Adopted from: Surway, R.A. and Jewett, J.W. (2014)

The student inside the train and the observer on the ground are in the same reference frame.

Galilean relativity (Example)

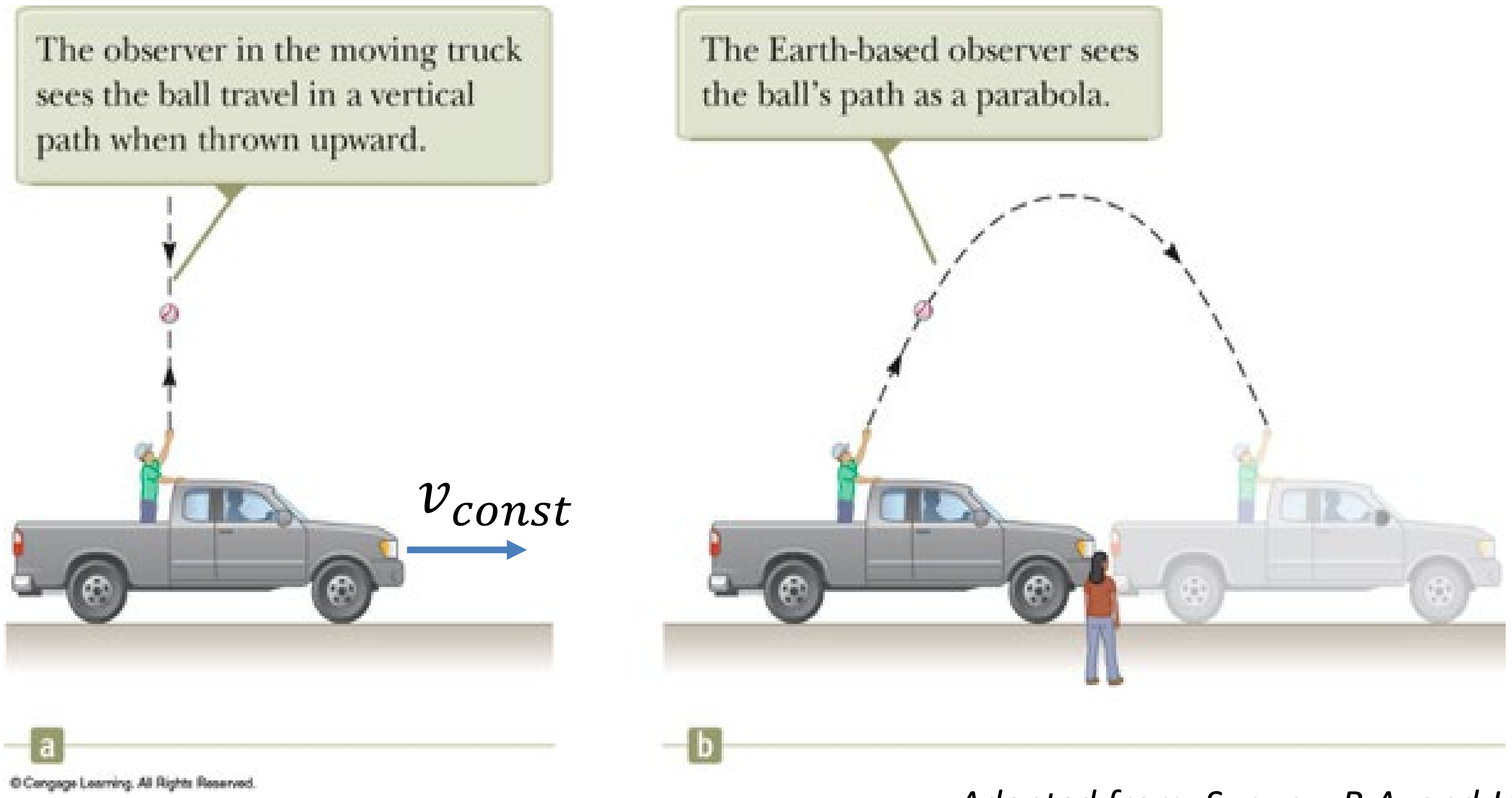


Adopted from: Surway, R.A. and Jewett, J.W. (2014)

If the truck move with constant speed, both the truck frame and the ground frame can be treated as inertial frame

✓ ✗

Galilean relativity (Example)



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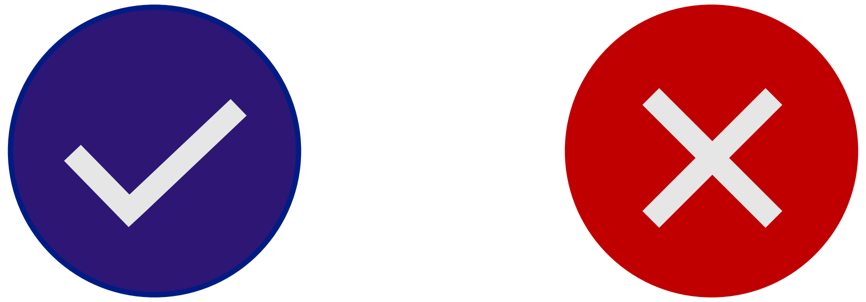
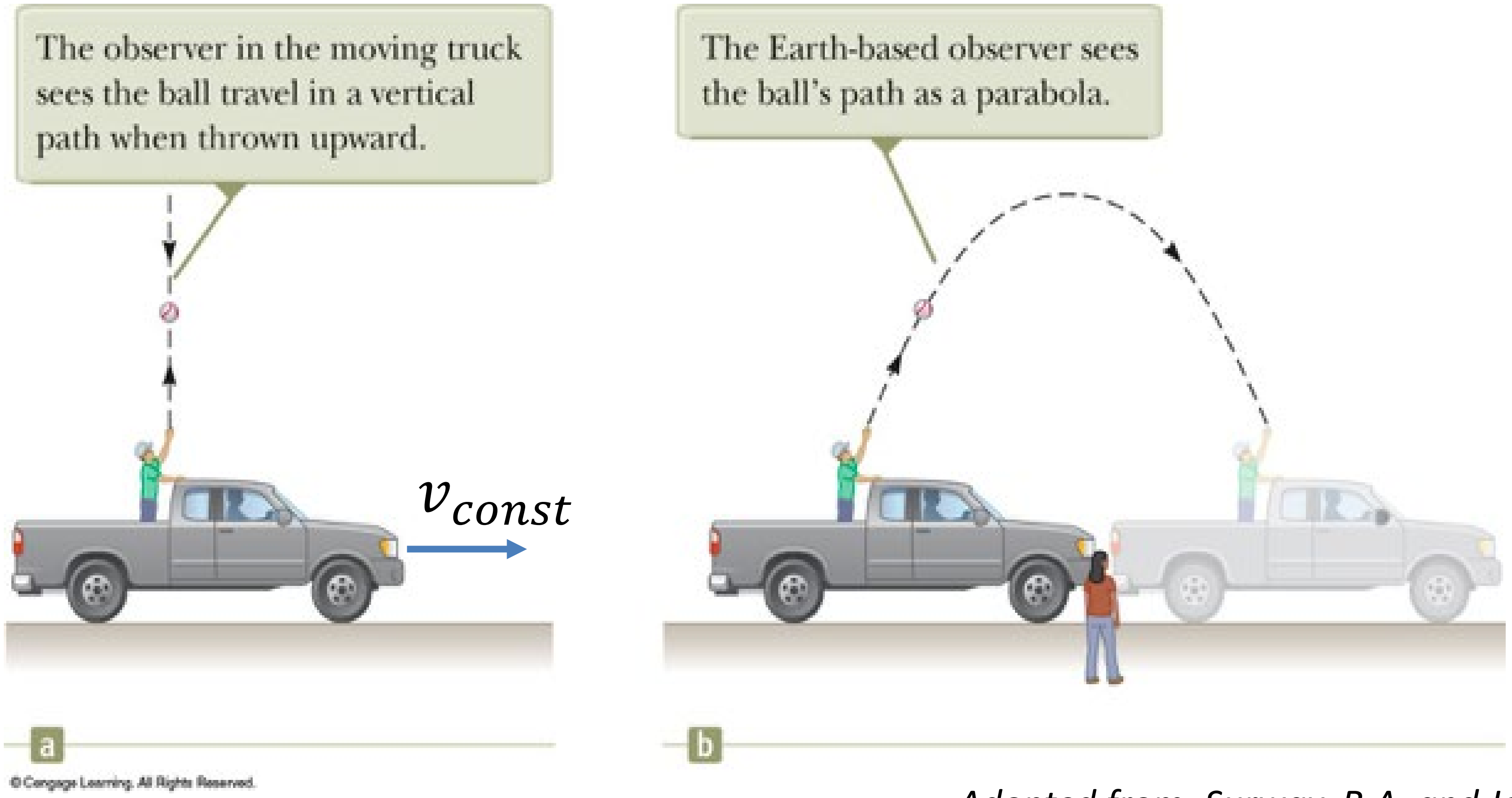
According to Galilean relativity, time is measure differently in the truck frame and the ground frame



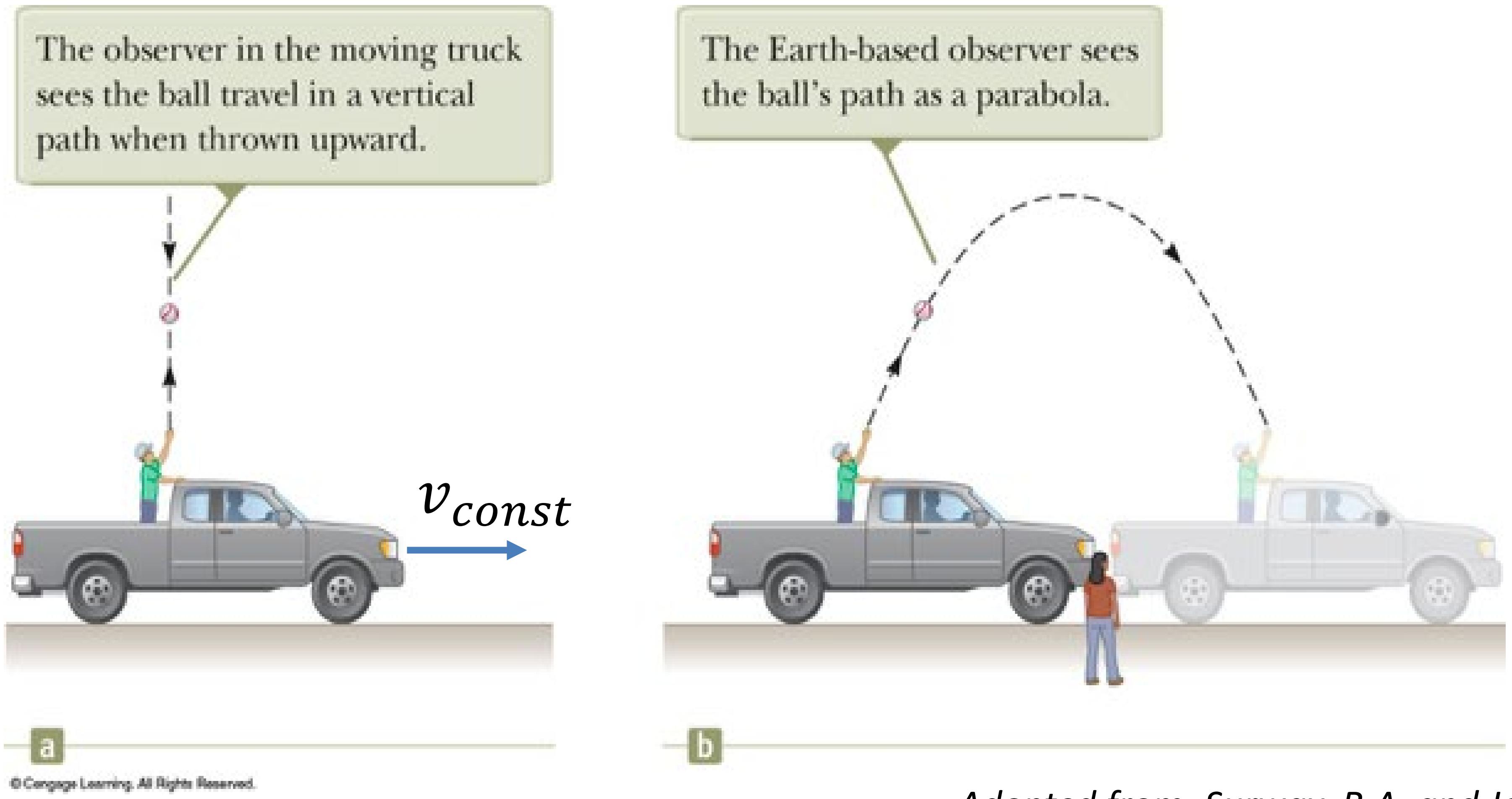
Galilean relativity (Example)



Adopted from: Surway, R.A. and Jewett, J.W. (2014)

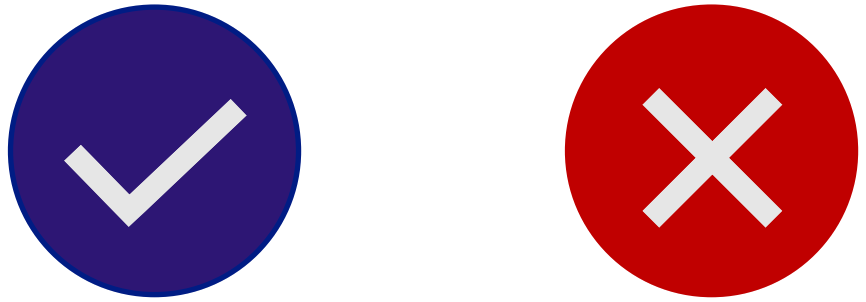
Velocity is a relative quantity because different observers may measure different values.

Galilean relativity (Example)



Adopted from: Surway, R.A. and Jewett, J.W. (2014)

The actual motion of the ball is different for each observer.

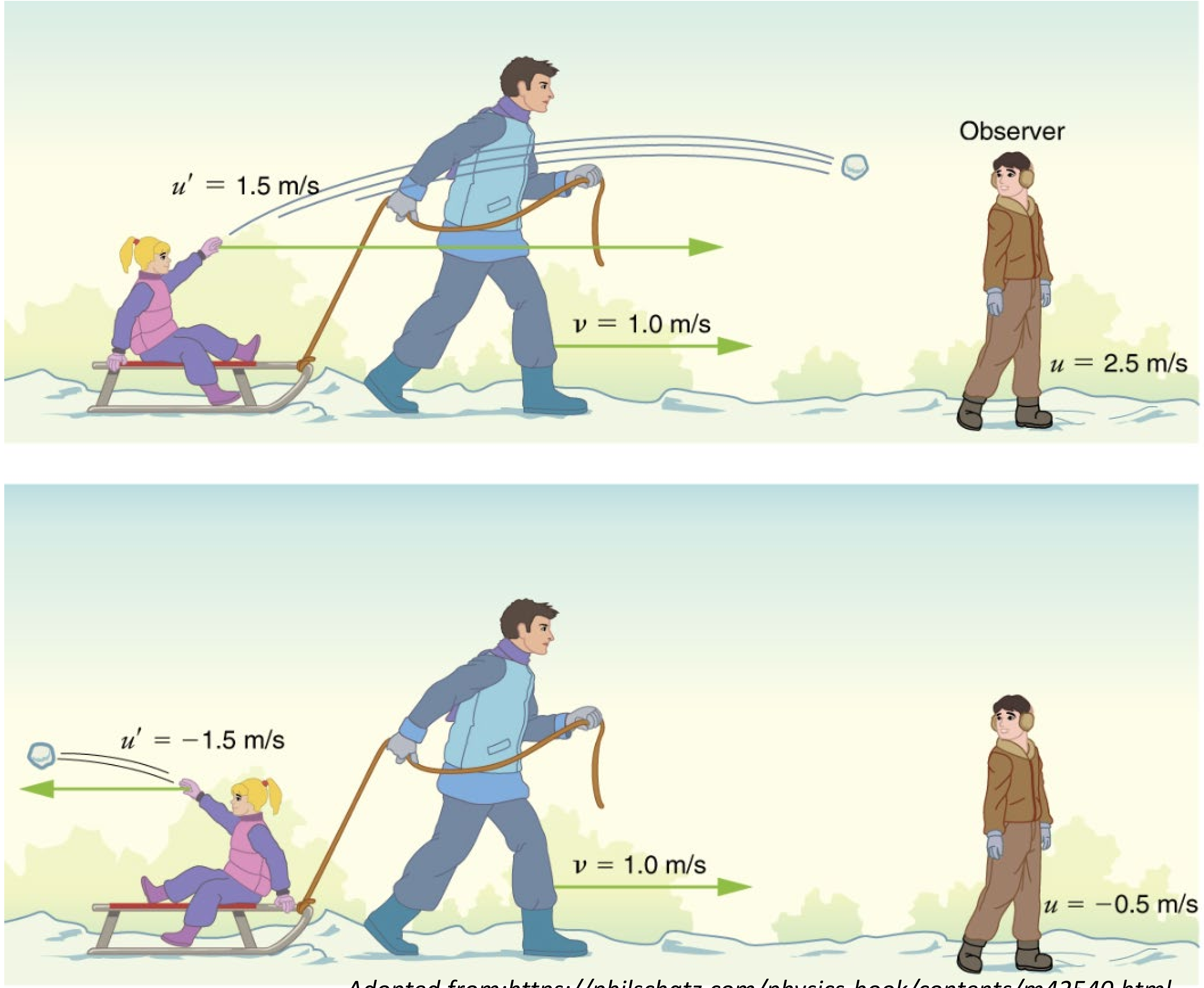


Galilean relativity (Example)

The moving (O’) frame

The rest (O) frame

Galilean Transformation (Classical Mechanics)



Adopted from:https://philschatz.com/physics-book/contents/m42540.html

Calculate the speed of light in the rest frame (Men observer)?

Adopted from:https://philschatz.com/physics-book/contents/m42540.html

Light Velocity (The Maxwell's equation)

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{B}}{\partial t}$$

Electromagnetic wave :

$$\frac{\partial^2 \vec{E}}{\partial x^2} = \mu_0 \varepsilon_0 \cdot \frac{\partial^2 \vec{E}}{\partial t^2}, \quad \frac{\partial^2 \vec{B}}{\partial x^2} = \mu_0 \varepsilon_0 \cdot \frac{\partial^2 \vec{B}}{\partial t^2}$$

Wave equation :

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{v^2} \cdot \frac{\partial^2 \psi}{\partial t^2}$$

Where ψ is wave function

Velocity of light is constant: $\boxed{v = 1/\sqrt{\mu_0 \varepsilon_0}}$ = 2.998x10^8 m/s

Calculate the speed of light in the rest frame (Men observer)?

Galilean transformation

$$u' = u + v$$

$$u' \neq u$$

$$c' \neq c$$

$$t' = t$$

Possible Problems!!!

- Principle of Relativity fails
- Maxwell's Equation fails
- Galilean transformation uncomplete
- Light (EM Wave) propagate through an Ether (Medium)

Problem!!!

1. ~~Principle of Relativity fails~~

2. ~~Maxwell's Equation fails~~

3. Galilean transformation uncomplete

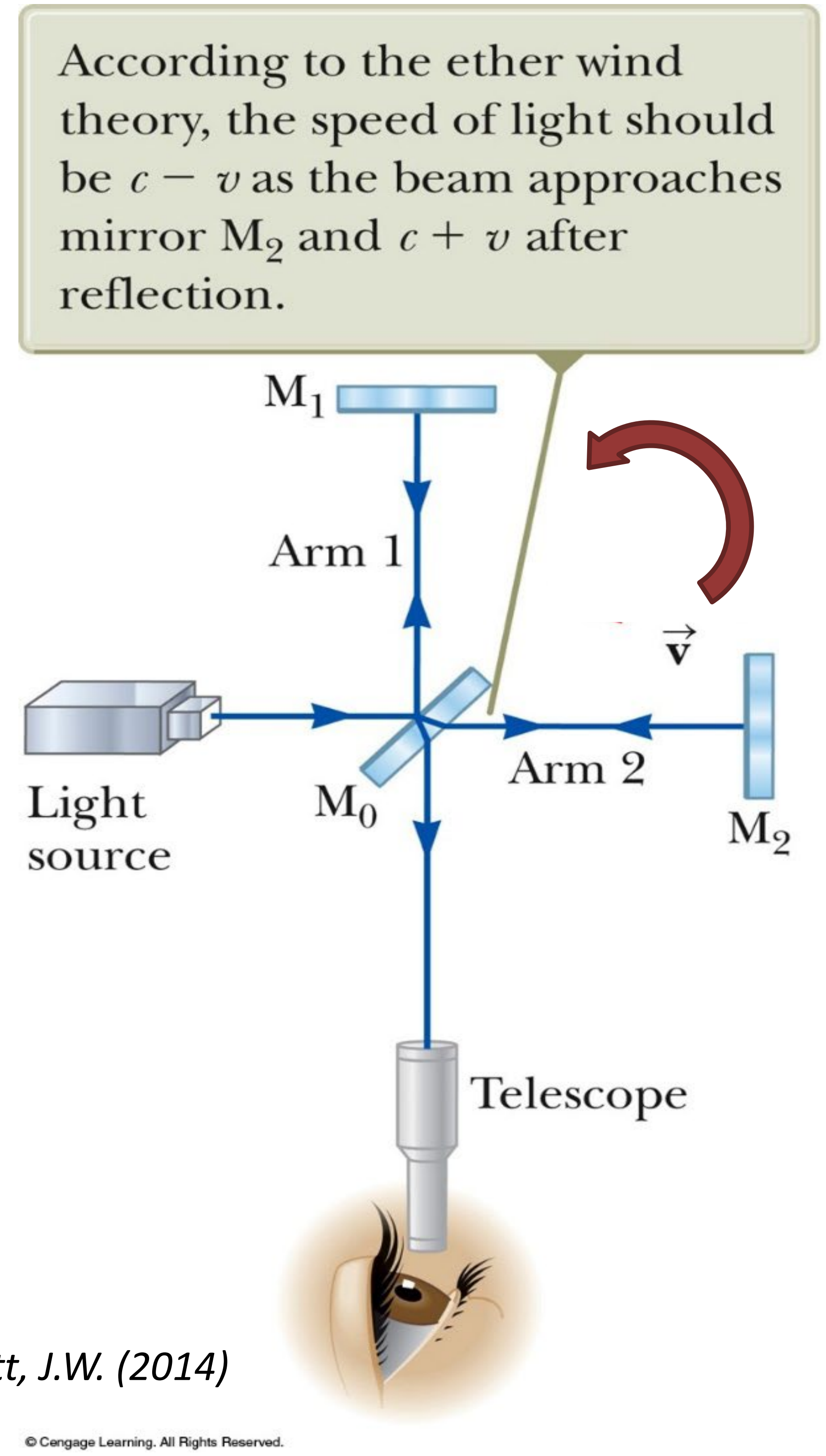
4. Light propagate in different medium (Ether)

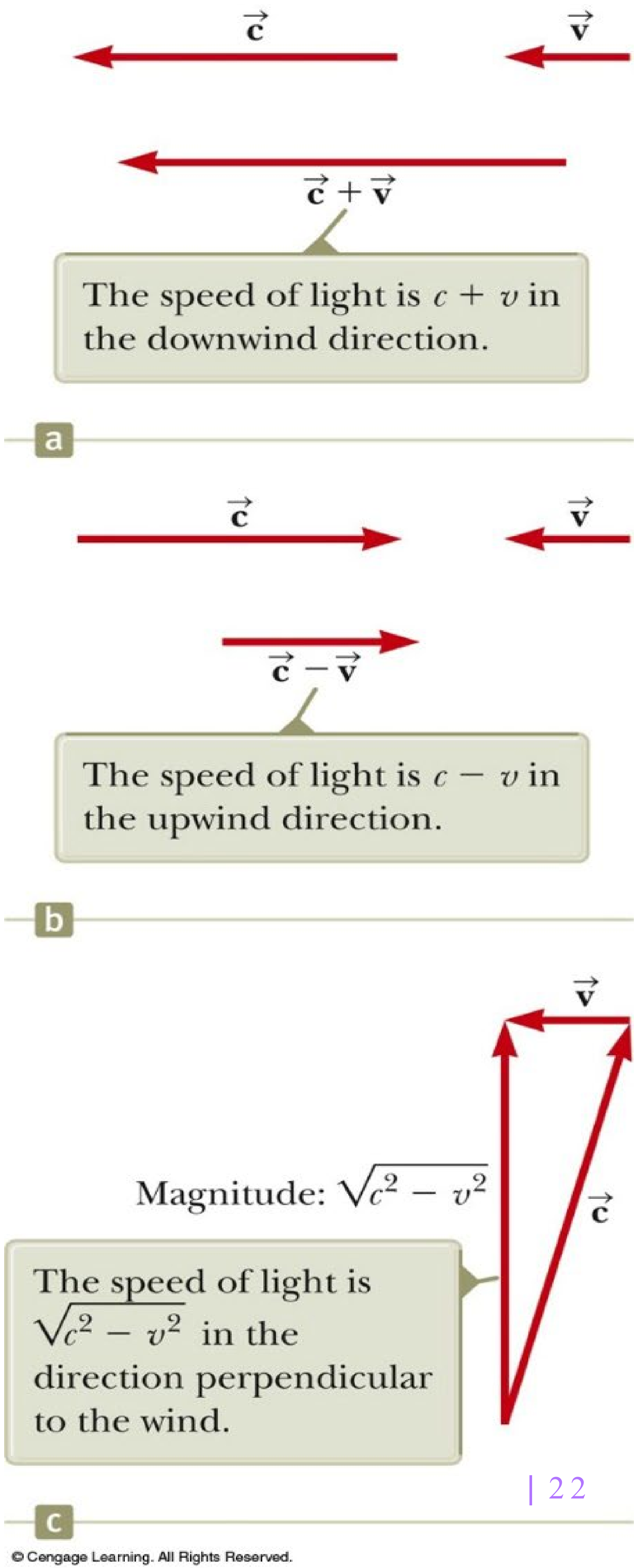
   Asyn: Michelson-Morley Experiment (in site)

Summary

- Describe the difference between inertial and non-inertial frame
- Could you explain the Galilean Principle?
- Are position, velocity and motion absolute or relative?

Michelson-Morley experiment





Adopted from: Surway, R.A. and Jewett, J.W. (2014)

Problem!!!

1. ~~Principle of Relativity fails~~
2. ~~Maxwell's Equation fails~~
3. Galilean transformation uncomplete
4. ~~Light propagate in different medium (Ether)~~

Einstein's special relativity

Resolves the contradiction between Galilean relativity and the fact that the speed of light is the same form for all observers.

- **The Einstein's postulates**

  The principle of relativity: The laws of physics have the same form in all inertial frames of reference.

  The constancy of the speed of light: the speed of light in a vacuum has the same value, $c = 3.00 \times 10^8$ m/s, in all inertial frames, regardless of the velocity of the observer or the velocity of the source emitting the light.

Take home messages

- Between time and the speed of light, Which one is always absolute? Explain why?

- What is the problem of the classical mechanics? Explain why?